1. Initialize an empty Sequential():

A. instead of passing all layers into a neural network constructor at once,
we can initialize an empty sequential neural network by "Sequential()" and append layers individually by ".add()" to build a layer-by-layer construction.

B. adding layers one by one significantly improve code readability and maintainability for DNN with lots of layers.

C. it also allows a more compact neural network description by using "for loop" to add multiple identical layers.

2. Deep Neural Network (DNN):

it features utilizing multiple hidden layers rather than just a single hidden layer to build a network topology.

3. The Advantages of Multiple Hidden Layers:

A. empirically better performance: theoretically, a single hidden layer with enough neurons is sufficient to allow a neural network to express basic functions,
however, empirical evidence shows that adding more hidden layers generally leads to build up a better-performing model in terms of "accuracy".

B. hierarchical combination of features: having multiple hidden layers enables a neural network to act like building blocks, combining features hierarchically.
lower hidden layers capture basic, simple elements from the raw input data, while higher hidden layers combine these basic and simple elements into increasing complex and larger-scale features.

C. increase levels of abstraction: through these multiple layers, a model can process information at the increasing abstraction levels,
on the other words, it gradually refines the micro-level, raw data (like exact numerical values) into the macro-level, higher-value concepts (like recognizing a specific object in an image or assessing overall neighborhood quality)

4. Definition of Overfitting in Deep Learning:

A. overfitting occurs when a model essentially memories the training dataset but fails to generalize its learning to unseen data,
and we can observe the steadily decreasing training error while the test error remains flat or stagnant.

B.